This means replacing narrative exposition with concrete visual actions and performable dialogue that convey the same story information through what can be seen and heard on screen, while following screenplay formatting conventions.

When training on dataset $\mathcal{D}$ to learn $P(S|X)$ in a single stage, gradients must simultaneously improve both narrative quality and format adherence, which can lead to conflicting optimization directions. Our proposed two-stage pipeline resolves this by introducing the Novel as an intermediate representation that decouples the learning objectives. Stage 1 optimizes for narrative generation independently, while Stage 2 optimizes for format conversion, eliminating gradient conflicts and simplifying each optimization problem.

### 3.2 The DSR Framework

Following the rationale for a decomposed strategy, we now detail the two-stage design of our DSR framework. As illustrated in Figure 1, this framework decouples creative narrative generation from stylistic format conversion by introducing novel-style prose as an intermediate representation, denoted as $N$. The complete generation process is formulated as a sequential sampling procedure:

**Stage 1: Outline-to-Novel Expansion with CoT.** The model learns to generate a Chain-of-Thought (CoT) analysis $I_c$, followed by the intermediate novel $N$, conditioned solely on the input $X$:

$$(\hat{I}_c, \hat{N}) \sim P(I_c, N \mid X; \theta_1)$$

**Stage 2: Novel-to-Screenplay Conversion.** The generated narrative prose $\hat{N}$ is then transformed into a structurally-correct screenplay $S$, guided by the original input $X$:

$$\hat{S} \sim P(S \mid \hat{N}, X)$$

This decomposition corresponds to a probabilistic model where the target distribution $P(S|X)$ is implicitly modeled through marginalizing over the latent narrative representation $N$.

The choice of novelistic prose as the intermediate representation is a central component of our framework's design. This choice is motivated by two observations about LLMs (OpenAI et al., 2024; Touvron et al., 2023). First, while LLMs are pretrained on vast corpora of narrative texts such as novels and stories, they have seen comparatively limited well-formatted screenplay data during training. Second, there exists a significant difference in information density between the two formats: novels can unfold plots with extensive descriptions and details, whereas screenplays must compress the same narrative into a much more concise format with strict structural constraints. Therefore, we introduce the novel as an intermediate representation, decomposing the generation task into two subtasks: first generating a novel from the outline, then converting the novel into a screenplay.

Notably, the intermediate novel is not a traditional literary novel. While a literary novel can describe characters' inner thoughts and use expressions like metaphors, the intermediate novel is specifically designed as a screenplay-oriented descriptive text. It serves as a practical guide for visual storytelling, with content constrained to observable actions and audible dialogue rather than abstract thoughts and emotions. This design bridges the gap between narrative generation and screenplay formatting. For instance, rather than stating a character's inner state as *"He was consumed by regret"*, it describes a performable moment: *"He stared at the cracked photograph, his jaw tight, before slowly closing his eyes"*.

The first stage **Outline-to-Novel Expansion** trains a model $M_{\theta_1}$ to perform both reasoning and creative generation. Its objective is to learn a mapping from the input brief $X$ to a structured output containing both a CoT analysis $I_c$ and the full novel prose $N$. By training the model to first generate a CoT, we encourage it to develop an internal reasoning process that enhances the quality and coherence of the subsequent novel generation. To achieve this, we train the model on a high-quality dataset $\mathcal{D}_{novel} = \{(X_i, (I_{c,i}, N_i))\}_{i=1}^{|\mathcal{D}_{novel}|}$.

Creating such a high-quality dataset is challenging. We have access to professionally written screenplays $S$, but not the original high-level briefs $X$ used to create them. We therefore develop a hybrid data synthesis strategy illustrated in Figure 2, combining the strengths of reverse and forward synthesis to construct our training pairs. The input $X$ is created via reverse synthesis to resemble a realistic writer's brief, while the target $N$ is created via forward synthesis to ensure high fidelity and narrative richness.

The foundation of our synthesis strategy is a high-quality, pre-processed screenplay corpus.[1] As

[1] See Appendix A.1 for detailed preprocessing steps.

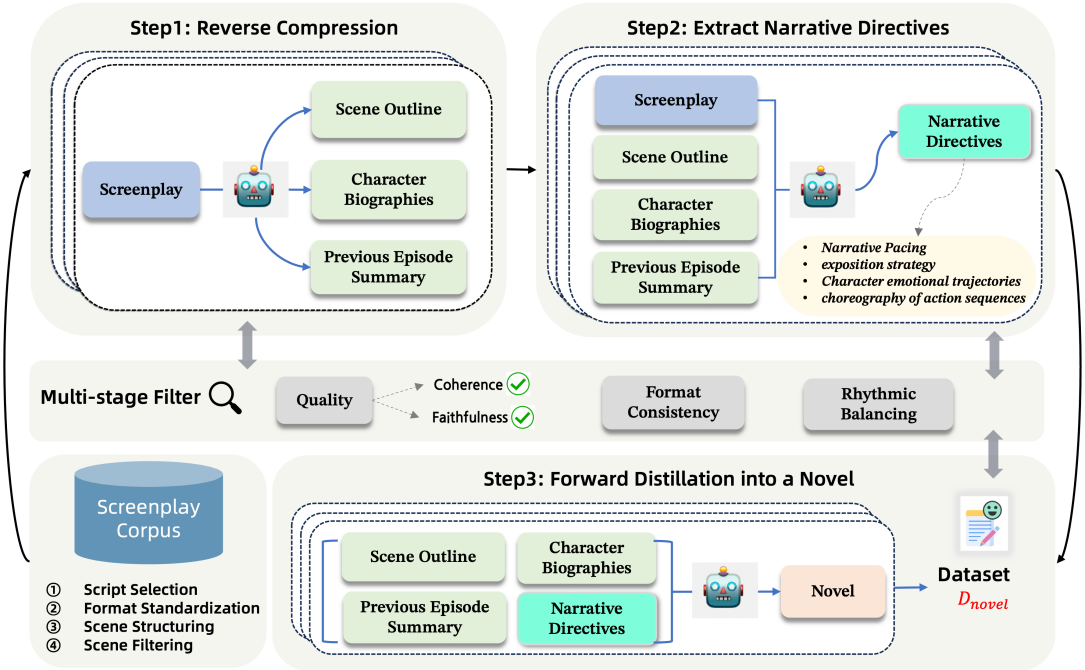


Figure 2: Workflow of the data synthesizing process. The goal of Step1 is to extract the core structural and narrative elements from a complete screenplay. Step2 aims to understand the "why" behind the screenplay. The objective of Step3 is to generate a new, long-form narrative (a novel) based on the structured information and narrative directives obtained in the previous steps.

shown in the lower-left corner of Figure 2, all raw scripts first pass through a standardized cleaning pipeline to ensure data quality and consistency. This resulting curated corpus contains over 200 series with more than 50000 scenes, serving as the source material for all subsequent synthesis tasks.

Based on this clean corpus, we perform the synthesis process in two main steps.

**Part A: Reverse Synthesis of Inputs ($X$) and Narrative Directives ($I_c$)** As depicted in Step 1: Reverse Compression of Figure 2, our process begins by taking a professional screenplay $S$ from the pre-processed corpus and reverse-engineering the corresponding input $X$ to simulate a writer's outline. However, reverse synthesis presents several key challenges. A primary difficulty lies in achieving the correct level of detail: the input must provide enough critical information (*e.g.*, character conflicts and plot turning points) without including so much that the task becomes unrealistic. Moreover, reliance on LLMs (*e.g.*, GPT-4) introduces the risk of hallucinations that create details inconsistent with the original screenplay, thereby reducing the quality of training data.

To overcome these difficulties, we reframe the task from simple "summarization" to "creative intent reconstruction" through iterative prompt engineering, focusing on extracting what happens in the story. Furthermore, all outputs undergo a rigorous human-in-the-loop review pipeline involving manual editing and quality filtering (Multi-stage Filter in Figure 2). The review process guarantees that the final input $X$ is accurate, representative of a realistic writer's outline, and maintains both coherence and faithfulness to the source material.

Concurrently, we move beyond the plot-level details of $X$ to capture the latent authorial strategy: the choices that dictate how the story is told. As shown in Step 2: Extract Narrative Directives of Figure 2, we analyze the source screenplay $S$ again to extract a set of parameters termed **Narrative Directives ($I_c$)**. These directives capture the underlying mechanics of storytelling, such as narrative pacing, the trajectory of character emotions, the choreography of action sequences, and the information disclosure strategy. This step provides a deeper layer of guidance that complements the structural information in $X$.

**Part B: Enhanced Forward Synthesis of Targets ($N$)** The goal of the forward synthesis phase is to generate our final training target: the novel $N$. While a forward approach guarantees consistency

with its inputs, naive forward synthesis that conditions only on the structured input $X$ ($X \rightarrow N$) often produces narratives that are logical but lack creative depth and professional quality. We therefore develop an enhanced forward synthesis process. As shown in Step 3: Forward Distillation into a Novel of Figure 2, we use a powerful teacher model (*e.g.*, Gemini, GPT-4) to generate the target novel $N$. Crucially, the generation is conditioned on both the structural input $X$ from Step 1 and the narrative directives $I_c$ from Step 2:

$$\hat{N} \sim M_{teacher}(X, I_c) \tag{2}$$

Here, $M_{teacher}$ denotes the "teacher model", a term from the teacher-student paradigm in machine learning, where a larger, highly capable model generates high-quality training data for a smaller, more specialized "student model". This enhanced process produces a target novel $N$ that is not only consistent with its input $X$ but also demonstrates narrative sophistication and design quality inspired by professional writing.

By combining reverse and forward synthesis, our final training pairs are structured as $(X_i, (I_{c,i}, N_i))$. The model is trained to sequentially generate the narrative directives $I_{c,i}$ followed by the novel $N_i$, conditioned on the input $X_i$. This design allows the model to learn both the reasoning process and the creative writing process.

With the high-quality dataset $\mathcal{D}_{novel}$, we apply supervised fine-tuning (SFT) to obtain $M_{\theta_1}$ from a base model. The training objective minimizes the standard negative log-likelihood loss over the entire target sequence, which is the concatenation of the narrative directives and the novel. Let $Y_i = (I_{c,i}, N_i)$ represent this full target sequence, where $Y_i = (y_{i,1}, \ldots, y_{i,T_i})$. The loss function is:

$$\begin{aligned} \mathcal{L}_{SFT}(\theta_1) &= -\frac{1}{|\mathcal{D}_{novel}|} \sum_{(X_i,Y_i)\in\mathcal{D}_{novel}} \log P(Y_i|X_i;\theta_1) \\ &= -\frac{1}{|\mathcal{D}_{novel}|} \sum_{(X_i,Y_i)\in\mathcal{D}_{novel}} \sum_{t=1}^{T_i} \log P(y_{i,t}|y_{i,<t}, X_i;\theta_1) \end{aligned} \tag{3}$$

Upon generating the rich narrative prose $N$ from Stage 1, the second stage, **Novel-to-Screenplay Conversion**, performs the stylistic transformation.

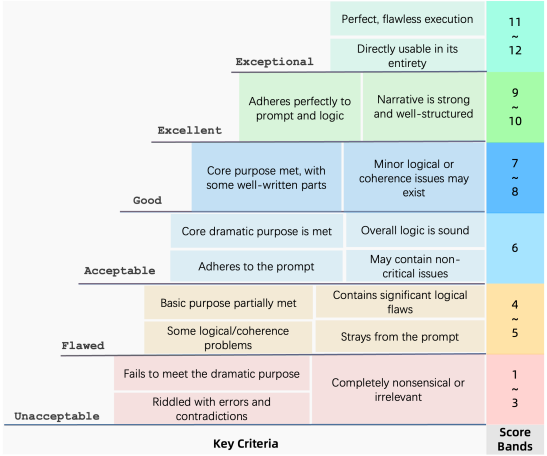


Figure 3: Holistic quality scoring rubric for screenplay evaluation. Each tier represents a quality level with corresponding score ranges and performance criteria.

This is an inference-only stage that requires no additional fine-tuning. We leverage the powerful in-context learning capabilities of a separate large-scale model, denoted as $M_{api}$ (*e.g.*, GPT-4), to approximate the distribution $P(S|N, X)$. The conversion is guided by a carefully engineered prompt $\pi$, which instructs the model to act as a professional screenwriter and provides clear formatting rules. The final screenplay $S$ is generated by sampling from this model:

$$S \sim M_{api}(\pi(N, X)) \tag{4}$$

The prompt $\pi(N, X)$ is carefully designed to provide clear context and instructions. It begins by establishing a specific persona through a role-playing directive, such as "You are a professional screenwriter," followed by a clear task definition. To ensure structural correctness, the prompt includes explicit formatting rules with examples of proper scene headings, action lines, and dialogue. Finally, the novelistic prose $N$ generated in Stage 1 is appended to these instructions as the input text for conversion.

In summary, our two-stage approach systematically develops both creative aspects of screenwriting. Stage 1 focuses on imaginative plot design and narrative development, while Stage 2 refines the output into proper screenplay format with appropriate character expression. By separating these tasks, we enable the model to produce scripts with both structural integrity and narrative quality.